

Machine Learning

Linear Regression- Week 4

Objectives

- Formulation of Linear Regression Equation
- Finding the best fit
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

Let's look at an example ...

Given, some data on housing prices in rural Portland, OR, our **task** is to predict the price of a house as a function of the size of their living areas.

Note, for simplicity we will derive the equation for one feature now. We will expand it to multiple features later.

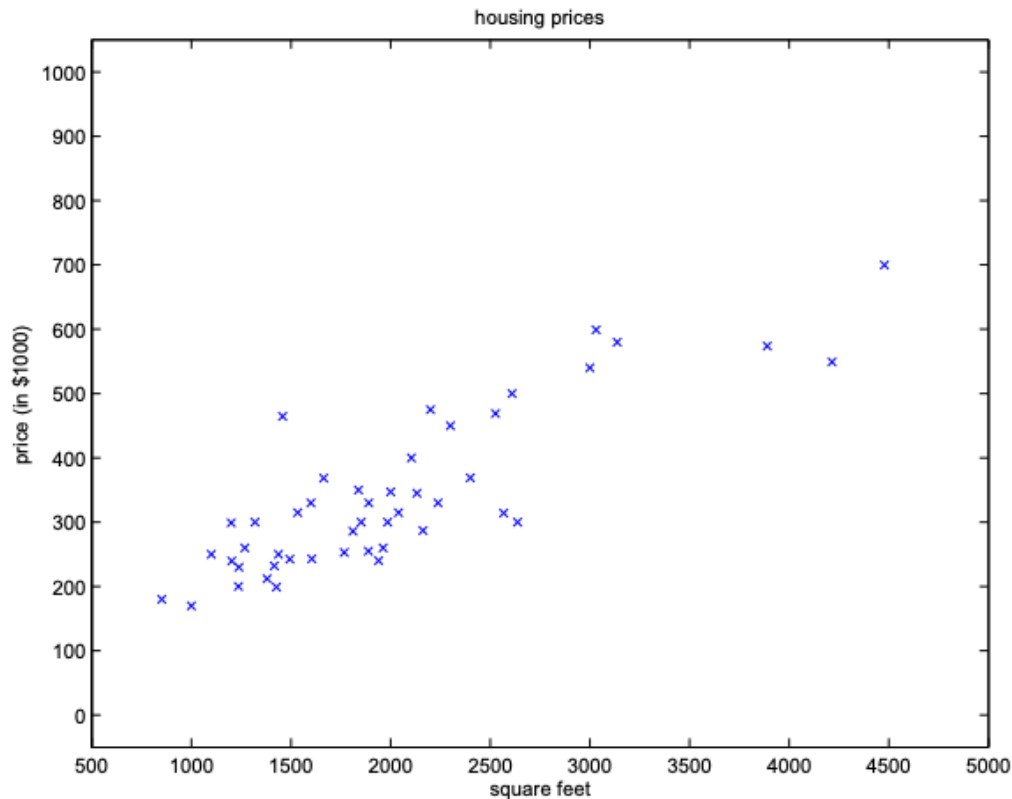
Living area (feet ²)	Price (1000\$)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

Table: Inventory of houses sold in Portland , OR

Let's look at an example ...

Let's plot our data

Living area (feet ²)	Price (1000\$s)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

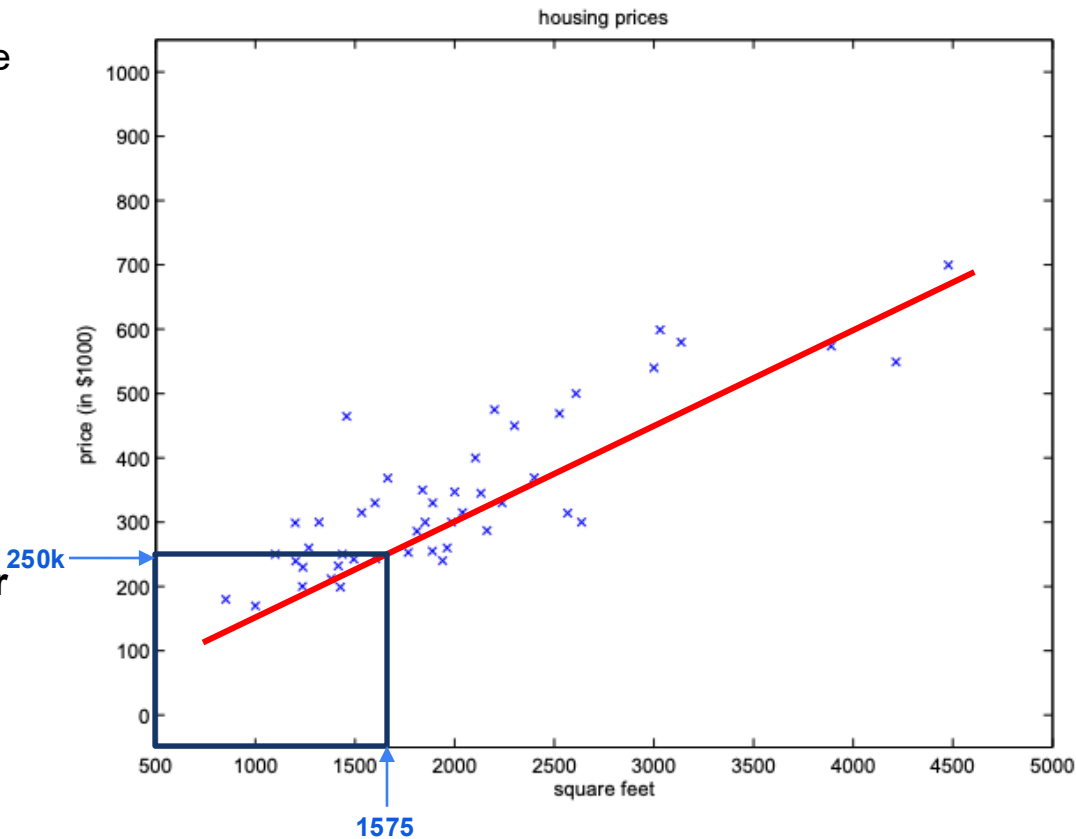


Let's look at an example ...

Now if we were to find the price of a house with 1575 sq.ft, we will fit a straight line to data and try to find the price corresponding to 1575 sq.ft

The machine learning model we create here are just mathematical representations of our input variable (square feet) and output variable (price in \$1000)

The ML model we develop here is a **linear regression** model



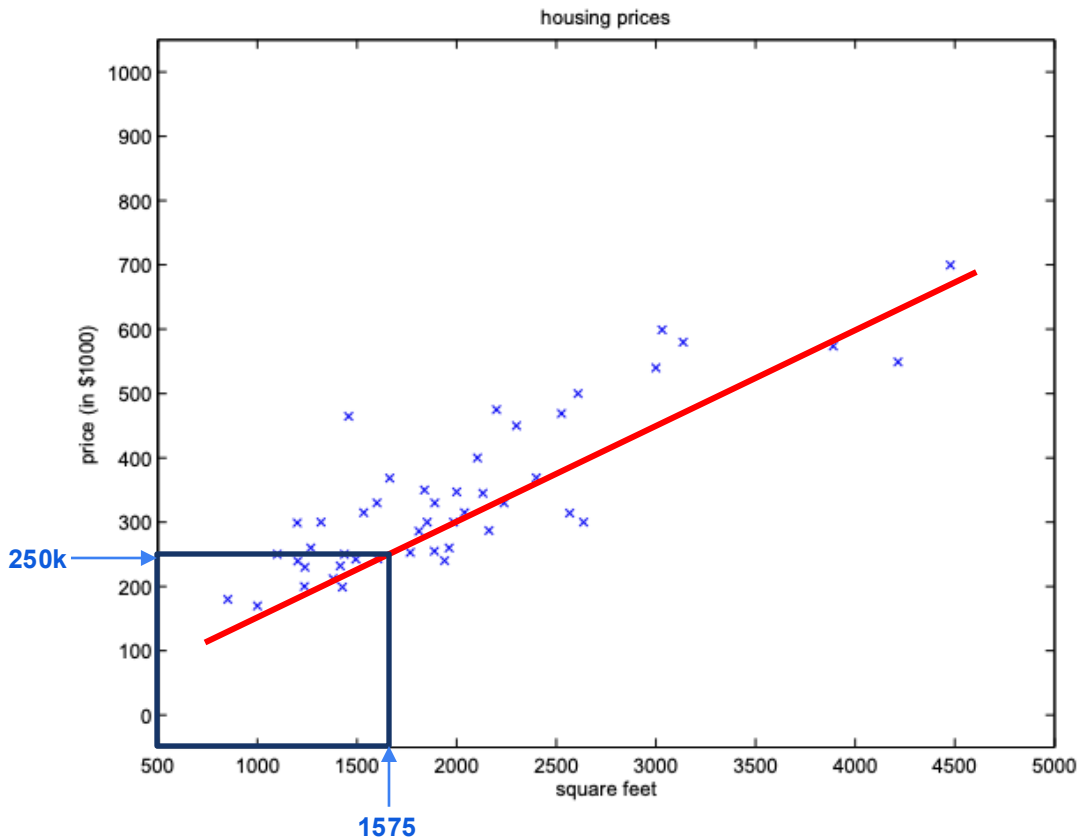
Let's look at an example ...

The ML model we develop here is a **linear regression** model:

Regression → since this predicts numbers and not distinct categories/classes

Linear → The relationship between input variables (or features), in our case square footage of the house and output (price) is linear, i.e. can be expressed in the form of a straight line:

$$y = mx + c$$



Terminology

x = input variable or features or attributes

y = output variable/ target

m = total number of training samples

In our case, m is equal to 47 samples

We denote a single training sample as

$(x^{(i)}, y^{(i)})$

Example, (x^1, y^1) is (2104, 400)

Living area (feet ²)	Price (1000\$)
$x^1 \rightarrow 2104$	$y^1 \rightarrow 400$
$x^2 \rightarrow 1600$	$y^2 \rightarrow 330$
2400	369
1416	232
3000	540
$\vdots$	$\vdots$

Linear Regression

The equation for linear regression, therefore, can be written as:

$$f_{w,b}(x) = w * x + b$$

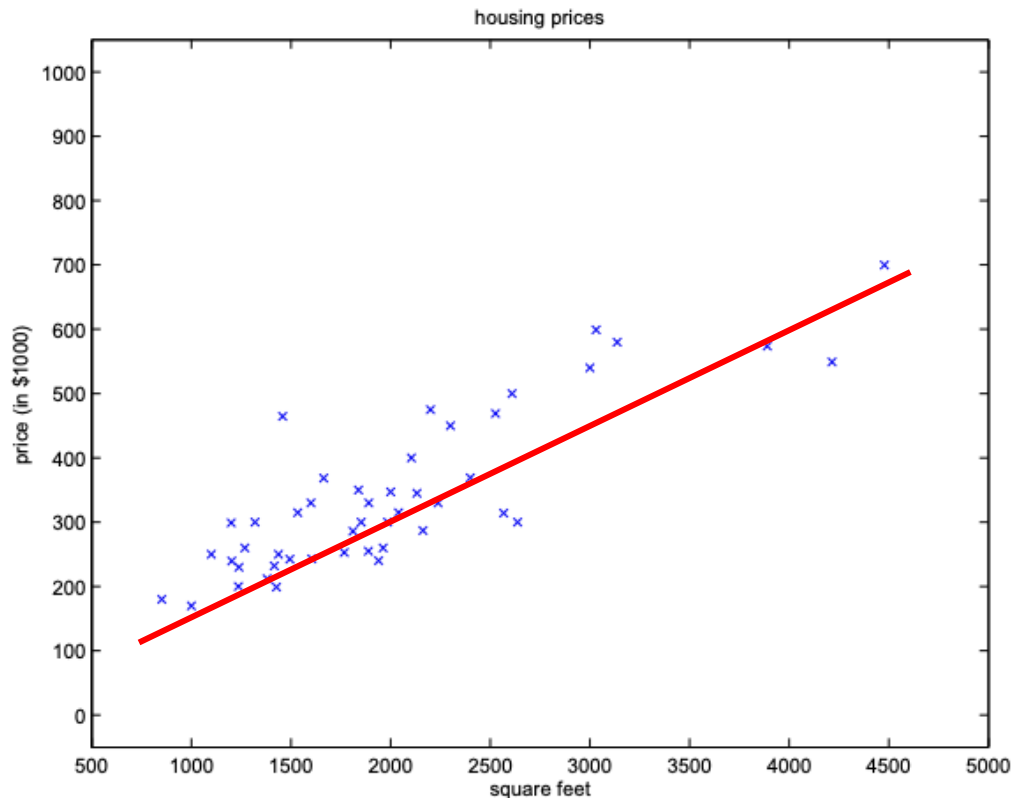
$w, b \rightarrow$ parameters/coefficients/weights and bias

$f(x)$ is the estimated value of y , in our case price(in \$1000)

We often denote $f(x)$ as $\hat{y}$

This is an example of **univariate linear regression**.

Univariate $\rightarrow$ “uni” or **one** & “variate” or **variable**



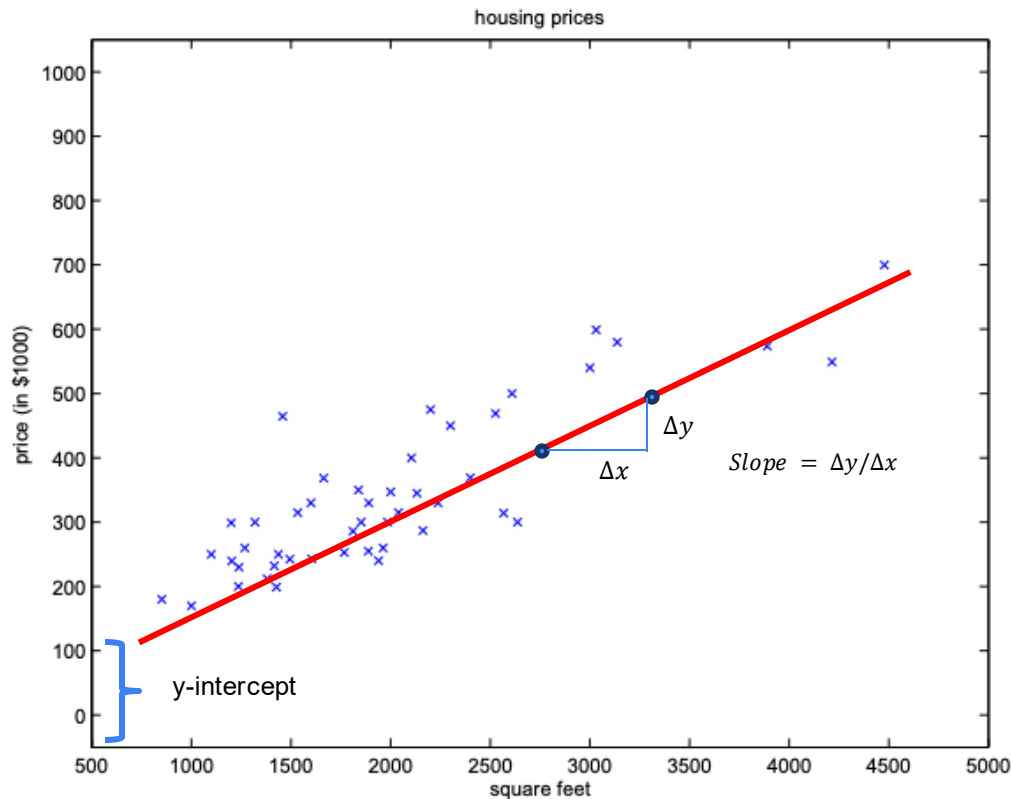
Linear Regression

The equation for linear regression,

$$f_{w,b}(x) = w * x + b$$

$w \rightarrow$ slope of the line

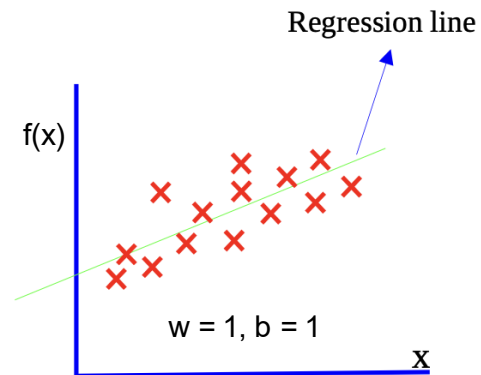
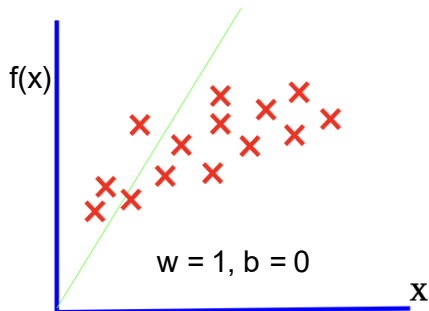
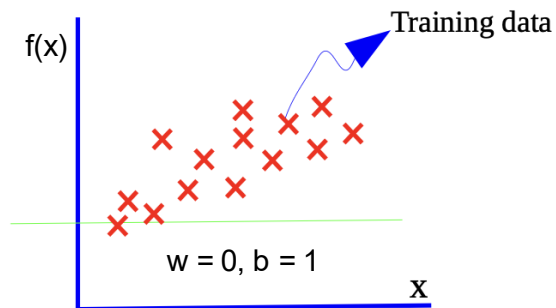
$b \rightarrow$ y-intercept



Significance of w and b

$$f_{w,b}(x) = w * x + b$$

Different values of w and b corresponds to different lines

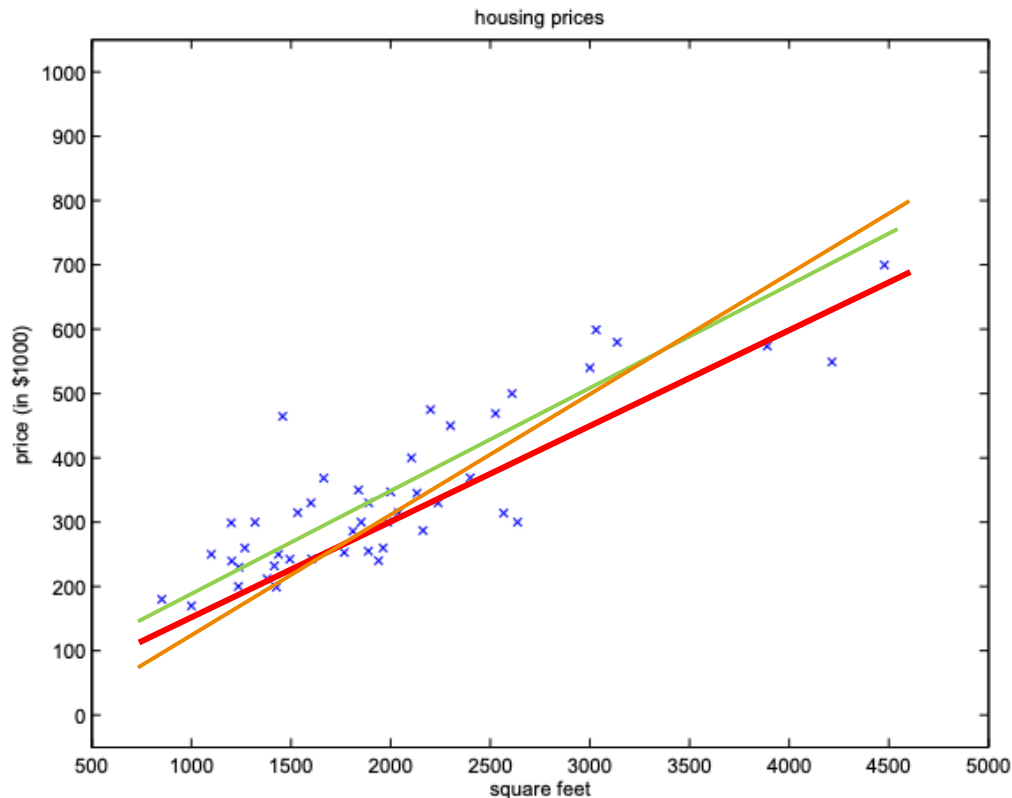


Finding the “best” fit...

For any data, we can get multiple lines or fits.

Out of these we need to select the one that passes through most of the training samples such that error between predicted values of y , i.e. $\hat{y}$ and actual value, y is minimized.

We will look at the error and cost function in next class



ANY QUESTIONS?